

Telco Customer Churn Prediction: Feature Engineering, Scaling and Model Comparison

Data Science Internship Program - Week 3 Assignment

1. Introduction

This project develops a machine-learning workflow for predicting customer churn using the IBM Telco Customer Churn dataset. The original dataset contains 7,043 customer records and 21 variables covering demographics, account information, subscribed services, charges, and the binary target variable *Churn*. The objective was to clean and prepare the data, engineer meaningful predictive features, compare feature-scaling methods, train multiple classification algorithms, and determine which model is best suited to the problem using both empirical results and conceptual reasoning.

2. Data Preparation

Initial inspection found no duplicate rows and no conventional null values. However, *TotalCharges* was stored as an object column and contained 11 blank-string values. These values were converted to missing values using numeric coercion. Because only 11 of 7,043 records (approximately 0.16%) were affected, the corresponding rows were removed rather than imputed. This avoided introducing artificial charge values while retaining virtually the entire dataset. The *customerID* field was also removed because it is a unique identifier rather than a meaningful behavioral, demographic, or account predictor. After cleaning, 7,032 customer records remained.

The target distribution contained 5,163 non-churn customers (73.42%) and 1,869 churn customers (26.58%). This moderate class imbalance means that accuracy alone is insufficient for evaluation, so precision, recall, F1-score, and ROC-AUC were also used.

3. Feature Engineering and Encoding

Engineered Feature	Purpose
AvgMonthlySpend	Estimates historical monthly spending from TotalCharges divided by tenure, capturing longer-term spending behavior
ContractCommitment	Maps month-to-month, one-year, and two-year contracts to increasing commitment levels because these categories have different churn rates
ServiceCount	Counts optional services used by a customer, representing the customer's level of integration with the provider's ecosystem
ChargeCommitmentInteraction	Combines monthly financial cost with contract commitment to capture cases where similar charges may have different churn implications

The target variable was encoded as No = 0 and Yes = 1. Remaining nominal categorical predictors were one-hot encoded so that arbitrary numerical rankings were not imposed on unordered categories. After encoding, the modeling dataset contained 35 columns and no missing values.

4. Feature Scaling and Train/Test Methodology

The prepared data was split into 80% training and 20% testing sets using stratification on *Churn*. This produced 5,625 training observations and 1,407 test observations while preserving the 73.42% / 26.58% class distribution in both sets. A fixed random state of 42 was used for reproducibility.

Two scaling methods were evaluated: StandardScaler and MinMaxScaler. StandardScaler centered the selected numerical variables around zero with unit standard deviation, while MinMaxScaler transformed the training values to the 0-1 range. Both scalers were fitted only on the training set and then applied to the test set. This order prevents test-set information from influencing preprocessing and avoids data leakage. StandardScaler was carried forward for Logistic Regression, K-Nearest Neighbors, and Support Vector Machine because these models can be sensitive to feature magnitude. Decision Tree and Random Forest were trained on the unscaled data because tree-based splits are scale-invariant.

5. Model Building and Evaluation

Five classifiers were trained: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest. In addition to evaluation on the held-out test set, 5-fold stratified cross-validation was performed on the training data using F1-score. F1 was selected for cross-validation because it balances precision and recall and is more informative than accuracy alone for the moderately imbalanced churn target.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Mean CV F1	CV Std
Logistic Regression	0.8038	0.6503	0.5668	0.6057	0.8363	0.5931	0.0223
Random Forest	0.7854	0.6169	0.5080	0.5572	0.8224	0.5663	0.0188
Support Vector Machine	0.7974	0.6540	0.5053	0.5701	0.7827	0.5779	0.0100
K-Nearest Neighbors	0.7633	0.5553	0.5508	0.5530	0.7798	0.5578	0.0103
Decision Tree	0.7122	0.4608	0.4866	0.4733	0.6400	0.5031	0.0236

Table 1. Test-set performance and 5-fold cross-validation results.

Logistic Regression achieved the strongest overall result. It obtained the highest accuracy (80.38%), F1-score (60.57%), ROC-AUC (83.63%), and mean cross-validation F1-score (59.31%). SVM produced the highest precision at 65.40%, but its recall, F1-score, and ROC-AUC were lower. Random Forest ranked second by ROC-AUC at 82.24%. Decision Tree performed worst overall, with an ROC-AUC of 64.00%, suggesting that a single unrestricted tree generalized poorly compared with the other approaches.

6. Algorithm Research and Dataset Suitability

Logistic Regression. Logistic Regression models the log-odds of a binary outcome as a linear combination of input variables. It is appropriate for binary churn prediction, works naturally with one-hot encoded predictors, and is comparatively interpretable. Its main structural limitation is that nonlinear effects must be represented explicitly through transformations or interaction features.

K-Nearest Neighbors. KNN classifies an observation according to nearby observations in feature space. Because its predictions depend directly on distances, scaling is essential. Its performance can deteriorate in higher-dimensional spaces, which is relevant because one-hot encoding expanded this dataset to 35 columns.

Support Vector Machine. SVM seeks a separating boundary with a maximum margin between classes and is sensitive to differences in feature scale. Kernel methods can model nonlinear decision boundaries, but the resulting model is generally less interpretable and can be more computationally expensive than Logistic Regression.

Decision Tree and Random Forest. Decision Trees recursively partition the feature space and naturally capture nonlinear relationships and interactions without requiring scaling. Individual trees are prone to overfitting. Random Forest reduces this variance by combining many trees trained on bootstrap samples and random feature subsets, usually improving generalization at the cost of reduced interpretability.

These characteristics relate directly to the Telco data. The dataset mixes continuous account variables with categorical service and contract variables, contains correlated behavioral information, and has a moderately imbalanced target. One-hot encoding created a representation that Logistic Regression could use effectively, while the engineered features exposed spending, service-use, commitment, and interaction effects. The weaker KNN result is consistent with the difficulty of distance-based learning in an expanded feature space. Random Forest substantially improved on the single Decision Tree, but the additional nonlinear flexibility did not outperform Logistic Regression in this experiment.

7. Selected Algorithm, Limitation and Conclusion

Logistic Regression was selected as the best-suited model among the five algorithms tested. The selection is supported by both the held-out test results and cross-validation: it led on accuracy, F1-score, ROC-AUC, and mean CV F1. The result is plausible because churn in this representation can be modeled effectively from additive effects involving tenure, charges, contract commitment, service usage, and the engineered interaction variables. The model also provides a practical interpretability advantage because feature coefficients can be examined to understand the direction of churn associations.

A limitation is Logistic Regression's linear log-odds structure. If the underlying churn process were dominated by complex nonlinear relationships and higher-order interactions that were not adequately represented through feature engineering, Random Forest could be preferable because it can learn such structures automatically and does not

require feature scaling. In the present experiment, however, Random Forest's lower F1-score and ROC-AUC do not justify choosing the more complex model.

Overall, the workflow produced a clean and reproducible churn-classification pipeline using deliberate data cleaning, four engineered features, appropriate categorical encoding, two scaling methods, stratified train/test evaluation, five classification algorithms, and 5-fold cross-validation. Logistic Regression provided the strongest balance of predictive performance, stability, and interpretability.

References

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